

Module: Machine Structure  
Academic year: 2025-2026

Sector: Mathematics

Semester: S1

## Homework

Note : ...../15

N° Inscription: ..... Name:.....Prenome :..... Group: ... Section: ....

### Note:

- Submit your homework between November 29th to December 10th. Each group must submit their Homework during their tutorial session (TD).
- Failure to submit the Homework on time will result in a 0/15.
- The answer must be on topic

### Exercise 1:

You receive three frames from an electromechanical measurement system:

- W1 (pure binary – 10 bits): 0011010110
- W2 (BCD – 12 bits): 0001 0000 0111
- W3 (Gray – 7 bits): 1101011

a. Convert W1, W2, and W3 to decimal values.

- W1= .....
- W2= .....
- W3= .....

b. Consider two adjacent values:  $N1 = 127$ ,  $N2 = 128$

- Give their 8-bit Gray code representations.
  - $N1 =$  .....
  - $N2 =$  .....

- Then state the property that becomes apparent when comparing these two representations:

.....  
.....  
.....  
.....  
.....  
.....

c. The following is claimed to be a valid BCD code: 0110 1010

- Show that this code is invalid in BCD.

.....  
.....  
.....  
.....

- List two common BCD coding mistakes, and explain how to detect them.

### **Exercise 2:**

Represent the following values using 12 bits:  $A = -356$ ,  $B = +219$ ,  $C = -1$ ,  $D = +0$

a. Give the representation of A, B, C, D in:

- In SVA :  $A = \dots\dots\dots$   
 $B = \dots\dots\dots$   
 $C = \dots\dots\dots$   
 $D = \dots\dots\dots$
- One's Complement (C1) :  
 $A = \dots\dots\dots$   
 $B = \dots\dots\dots$   
 $C = \dots\dots\dots$   
 $D = \dots\dots\dots$
- Two's Complement (C2)  
 $A = \dots\dots\dots$   
 $B = \dots\dots\dots$   
 $C = \dots\dots\dots$   
 $D = \dots\dots\dots$

b. Structural analysis

1. Demonstrate why C1 provides two different representations of zero.

.....

.....

.....

.....

2. Give an example of an addition in C1 that produces an incorrect result, even when no visible error occurs.

.....

.....

.....

.....

- c. Calculate in 12-bit two's complement:  $(A + B) - C$ , specify if there is overflow. Justify your answer and give the decimal value.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

d. We add 8 bits  $A = a_7 \dots a_0$  to 8 bits  $B = b_7 \dots b_0$ , obtaining a result  $S = s_7 \dots s_0$  and a carry  $r$ .

1. If  $A$  and  $B$  have opposite signs, can there be an overflow?

.....

.....

.....

.....

2. If  $A$  and  $B$  have the same sign, what values can  $s_7$  and  $r$  take? In which case(s) does there be an overflow?

.....

.....

.....

.....

.....

.....

.....

.....

3. We are interested in the two most significant carries:  $r$  and  $r_6$  (which come from the addition of  $a_6$ ,  $b_6$ , and  $r_5$ , and are added to  $a_7$  and  $b_7$  to give  $s_7$  and  $r$ ). Referring back to the previous question, give the expression for the overflow as a function of  $r_6$  and  $r$ .

.....

.....

.....

.....

.....

**Exercise 3:** We use a simplified floating-point format: **1 sign bit, 6 exponent bits and 9 mantissa bits**

We consider the real numbers:  **$X = 37.8125$ ,  $Y = -0.00390625$ ,  $Z = 1024.75$**

**Part A:** Encode the numbers  $X$ ,  $Y$  and  $Z$  in this floating-point format and give the result in hexadecimal

1. Representation of  $X$  :

.....

.....

.....

.....

.....

.....

.....

.....

- .....
- .....
- .....
- .....

[illegible][illegible]

Part B :

[illegible]